

Example 2.2

$$X_t = \frac{1}{2} X_{t-1} + Z_t$$

$$\Rightarrow X_t - \frac{1}{2} X_{t-1} = Z_t$$

$$\Rightarrow \left(1 - \frac{B}{2}\right) X_t = Z_t$$

$$X_t = \left(1 - \frac{B}{2}\right)^{-1} Z_t \quad (*)$$

$$\text{Let } f(B) = \left(1 - \frac{B}{2}\right)^{-1}$$

$$f'(B) = \left(\frac{1}{2}\right) \left(1 - \frac{B}{2}\right)^{-2}$$

$$f''(B) = 2 \left(\frac{1}{2}\right)^2 \left(1 - \frac{B}{2}\right)^{-3}$$

$$f'''(B) = 3! \left(\frac{1}{2}\right)^3 \left(1 - \frac{B}{2}\right)^{-4}$$

If $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$

$$\text{then } f^{(k)}(B) = k! \left(\frac{1}{2}\right)^k \left(1 - \frac{B}{2}\right)^{-(k+1)}$$

Taylor series expansion of $f(B)$ about 0 is

$$f(B) = f(0) + f^{(1)}(0) B + \frac{f^{(2)}(0) B^2}{2!} + \dots$$

$$= \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j B^j$$

$$\begin{aligned} \therefore X_t &= \sum_{j=0}^{\infty} \left(\frac{B}{2}\right)^j Z_t \\ \text{in } (*) &= \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j Z_{t-j} \end{aligned}$$

$$\begin{aligned} E(X_t) &= E\left(\sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j Z_{t-j}\right) \\ &= \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j E(Z_{t-j}) \end{aligned}$$

$$= 0 \quad \because E(Z_t) = 0 \text{ for all } t.$$

$$\text{Var}(X_t) = \text{Var}\left[\sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j Z_{t-j}\right] \quad (**)$$

If $Z_1, \dots, Z_n$ are independent
and $a_1, \dots, a_n$ are constants

$$\text{Var}\left(\sum_{j=0}^{\infty} a_j Z_j\right) = \sum_{j=0}^{\infty} a_j^2 \text{Var}(Z_j)$$

in (**)

$$= \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^{2j} \text{Var}(Z_{t-j})$$

$$= \sigma_Z^2 \sum_{j=0}^{\infty} \left(\frac{1}{4}\right)^j \quad \left. \begin{array}{l} \text{sum of geometric} \\ \text{series} \\ a=1, r=\frac{1}{4} \end{array} \right\}$$

$$= \sigma_Z^2 \frac{1}{1-\frac{1}{4}}$$

$$= \frac{4}{3} \sigma_Z^2.$$

Suppose $h > 0$

Consider :

$$\begin{aligned}\text{Cov}(X_t, X_{t+h}) &= \mathbb{E}(X_t X_{t+h}) - \mathbb{E}(X_t) \mathbb{E}(X_{t+h}) \\&= \mathbb{E}(X_t X_{t+h}) \\&= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j Z_{t-j} \right) \left(\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k Z_{t+h-k} \right) \right] \\&= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^j \left(\frac{1}{2}\right)^k \mathbb{E}(Z_{t-j} Z_{t+h-k})\end{aligned}$$

$\{Z_t\}$ terms are independent

$$\begin{aligned}\mathbb{E}(Z_{t-j} Z_{t+h-k}) &= 0 \\&= \mathbb{E}(Z_{t-j}) \mathbb{E}(Z_{t+h-k})\end{aligned}$$

where $j \neq h-k$

Non-zero where $-j = h-k$

$$\Rightarrow k = j+h$$

$$\therefore \text{Cov}(X_t, X_{t+h}) = \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j \left(\frac{1}{2}\right)^{j+h} \mathbb{E}(Z_{t-j}^2)$$

$$\mathbb{E}(Z_{t-j}^2) = \sigma_z^2$$

Because $\mathbb{E}(Z_{t-j}) = 0$
and $\therefore \text{Var}(Z_{t-j}) = \mathbb{E}(Z_{t-j}^2)$

$$\Rightarrow \text{Cov}(X_t, X_{t+h}) = \sigma_z^2 \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^{2j+h}$$

} Geom series

$$= \left(\frac{1}{2}\right)^h \sigma_z^2 \sum_{j=0}^{\infty} \left(\frac{1}{4}\right)^j$$

$$= \frac{4\sigma_z^2}{3 \cdot 2^h}$$

where $h < 0$ $\text{Cov}(X_t, X_{t+h}) = \frac{4\sigma_z^2}{3 \cdot 2^{|h|}}$

$\therefore \gamma(h) = \text{Cov}(X_t, X_{t+h})$ does not depend on t .

$\therefore$ Process is (weakly) stationary.